

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0980

Somerville 138 kV
May 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0980. The Interconnection Customer proposes a 84.3 MW natural gas combined cycle facility interconnecting at the Somerville 138 kV substation in the Pennsylvania area. The request was received on April 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 403 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$41,139,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kilarc / Thermas 138.0 kV	Conemaugh Hydro Plant / Thermas 138.0 kV Line	176.2	177	5.9
St Croix Falls / Emily 3 138.0 kV Line	Burwell Substation / P.T.S 345.0 kV	148.1	199	10.3
Chase Hill / Emily 3 138.0 kV	Burwell / P.T.S 345.0 kV	146.2	211	10.3
Mcintosh Combined Cycle Facility / St Croix Falls 138.0 kV	Burwell / P.T.S 345.0 kV	124.3	222	10.3
Placer / Stinson Substation 138.0 kV	Southeast Vicksburg / Allegheny Energy Units 3 4 & 5 500.0 kV	122.4	331	6.1
Wilson 1&2 / Mcintosh Combined Cycle Facility 138.0 kV	Burwell / P.T.S 345.0 kV	119.2	227	10.3
Chase Hill / Jones Bluff 138.0 kV L	Sugar Hill / Allegheny Enrgy Units 3 4 & 5 345.0 kV line	118.7	188	12.2
Eklutna / Allegheny Hydro No 8 345.0 kV	Caldwell / Piney Crk Project 345.0 kV	109.7	1,188	5.3
Placer 138 kV/1 kV Transformer	Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500.0 kV	107.2	378	6.1
Firstenergy Springdale Power Sta / Jones Bluff 138.0 kV	Sugar Hill / Allegheny Energy Units 3 4 & 5 345.0 kV	105.9	194	12.2
N Gat Riverside Alte / P.T.S 138.0 kV Line	Burwell / P.T.S 345.0 kV	105.5	245	10.9
N Gate Riverside Alte / Delco 138.0 kV	Burwell / P.T.S 345.0 kV	105.2	233	10.9
Pacolet Diesel Generation Facility / Grove 138.0 kV	Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500.0 kV Line	104.8	391	8.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
North Waverly / Grove 138.0 kV	Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500.0 kV L	104.2	405	8.6
North Waverly / Firstenergy Springdale Power Sta 138.0 kV Line	Allegheny Energy Units 3 4 & 5 Substation / Southeast Vicksburg Substation 500.0 kV	104.0	408	8.6
Blackrock / Eklutna 345.0 kV	Caldwell Substation / Piney Crk Project 345.0 kV	101.3	1,380	5.1
Little Chute / Talenenergy Martins Creek Llc Harrisburg 345.0 kV	Fort Humbug / Coronet 500.0 kV	100.9	2,249	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kilarc / Thermas 138.0 kV	Rebuild line with higher-capacity conductor	\$3,845,000	100% (\$3,845,000)
St Croix Falls / Emily 3 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,977,000	100% (\$2,977,000)
Chase Hill / Emily 3 138.0 kV	Rebuild line with higher-capacity conductor	\$4,539,000	100% (\$4,539,000)
St Roix Falls / Mcintosh Combined Cycle Facility 138.0 kV	Rebuild line with higher-capacity conductor	\$5,509,000	100% (\$5,509,000)
Placer / Stnson 138.0 kV	Rebuild line with higher-capacity conductor	\$1,998,000	20% (\$410,000)
Wilson 1&2 / Mcintosh Combined Cycle Facility 138.0 kV	Rebuild line with higher-capacity conductor	\$4,779,000	100% (\$4,779,000)
Chase Hill / Jones Bluff 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,166,000	100% (\$2,166,000)
Eklutna / Allegheny Hydro No 8 345.0 kV	Reconductor line	\$34,436,000	18% (\$6,192,000)
Placer 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$3,378,000	20% (\$692,000)
Firstenergy Springdale Power Sta / Jones Bluff 138.0 kV	Reconductor line	\$2,315,000	100% (\$2,315,000)
N Gat Riverside Alte / P.T.S 138.0 kV Line	Reconductor line	\$481,000	20% (\$94,000)
N Gate Riverside Alte / Delco 138.0 kV	Reconductor line	\$536,000	20% (\$105,000)
Pacolet Diesel Generation Facility / Grove 138.0 kV	Reconductor line	\$2,105,000	100% (\$2,105,000)
North Waverly / Grove 138.0 kV	Reconductor line	\$2,095,000	100% (\$2,095,000)
Firstenergy Springdale Power Station / N Waverly Substation 138.0 kV	Reconductor line	\$1,417,000	100% (\$1,417,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blackrock / Eklutna 345.0 kV	Reconductor line	\$21,588,000	8% (\$1,802,000)
Little Chte / Talenenergy Martins Crk Llc Harrisburg 345.0 kV	Reconductor line	\$8,317,000	1% (\$97,000)
	Total allocated to project		\$41,139,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.